

# ZBIGNIEW (BILL) OLIK

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## PERSONAL SUMMARY

Dynamic Senior Quantitative Developer with extensive experience at Millennium Partners, specializing in pricing implementation and risk analytics for diverse asset classes. Proven expertise in C++ and Java, complemented by strong leadership in model development and optimization. Recognized for enhancing performance and delivering impactful solutions in high-pressure environments.

## SKILLS

- Interest Rate Products
- FX Derivatives
- Commodities
- Credit Derivatives
- Hedging
- Monte Carlo Simulation
- Finite Differences
- Numerical Methods
- Architecture & design
- Code refactoring
- Numerical code optimization
- Design patterns
- Full product lifecycle
- PnL Explain
- Factor Analysis
- Stress Testing
- Model Drawdown Behavior
- Portfolio Theory
- Methodologies and Regulations
- Linear & Nonlinear Regression
- Time Series
- PCA
- C++ 17
- Boost
- Java
- Python
- Groovy
- CUDA
- Matlab
- R
- SAS
- Perl
- Excel
- VBA
- C++ programming
- Java programming
- Model development
- Risk analytics
- Pricing implementation
- Library design
- Full-stack development
- SQL and NoSQL databases

## EXPERIENCE

### SENIOR QUANTITATIVE DEVELOPER, FIC | 07/2014 to Current Millennium Partners - New York, NY

- Worked on various asset classes and technologies supporting Fixed Income Business
- Worked both on Model Development and System Implementation
- Implemented pricing, risk analytics, and technology components for IR, FX and Credit, and Commodity products
- This included model development, analysis, and implementation for real time pricing and risk batch applications
- Provided front office support to various traders and risk and middleware teams
- Technologies include Java, C++ 11/14, CUDA, Kafka, Solace, gRPC, MongoDB, MS SQL, Python, Linux shell scripting
- Rates: Implemented Beta management Tool for the SABR model
- Ran PnL impact across portfolios for Betas
- Caps/Floors: implemented model for real time pricing, greeks, risk batch, scenarios
- Forward Volatility Agreements: implemented model for real time pricing, greeks, risk batch, pillar scenarios
- Vega Hedging: created replicating swaption vol attribution for Midcurve swaptions, FVAs, CMS Spread options
- Volatility Converter: Wrote framework to recalculate market vols using different basis

- of funding and projection
- Optimized implementation of CRR Binomial Tree price in Java and C++ reducing calculation times
- FX: Refactored FX Vanilla and Digital Options real time pricing framework
- Added premium adjustment and adapted Greeks to models, which included PnL Analysis across all currency pairs across various trader groups
- Dual Digital Options: implemented copula model for real time pricing, greeks, risk batch, pillar scenarios
- Volatility Interpolation: developed FX volatility interpolation framework in C++ to improve performance
- Commodities: Implemented future Option Pricing T-1 settles from LME Metal Quotes
- Automated Volatility surface generation for Commodity Option using JPM surfaces
- Supported Various Futures, Future Options, Asian Price Options
- Supported Curve Shaping for various commodity curves
- Credit: Supported Bond Pricing, Credit ETFs, Credit Index Options, Tranche Pricing
- Tech: Market Data – rewrote interface to access market data for live and EOD Market Data for various asset classes
- C++ Analytics Library development
- Ported various Java components to C++
- CUDA: Implemented pricing framework leveraging parallel computations, including parallel CRR Binomial model
- Led design and implementation of Mongo DB sharded cluster
- Wrote Various services in Python
- Provided technical guidance to team members
- Wrote JNI layer to call C++ code from Java
- Created universal jar file to deploy both on Windows and Linux and not required external shared libraries

#### **SENIOR RISK ANALYST & DEVELOPER, RISK MANAGEMENT TEAM | 10/2012 to 04/2014**

##### **Two Sigma Investments - New York, NY**

- Led the development of a new Risk Reporting and Alerting Platform for use by Risk and Compliance teams for use as an independent control function; this included daily metrics for region / portfolio / risk factors
- Platforms include Java, Groovy, and SQL Database
- Frontend was Excel as well as Custom Report Web UI
- Developed and presented various reports to senior Risk/Legal/Compliance leadership. This included Daily Ex Post PnL Attribution Analysis for Models and Portfolio for different asset classes
- Developed Framework for Mining Large Forecast Data sets for daily forecasts
- Researched and Developed methodology for measuring model drawdown behavior
- Development Lead of a new Model Release and Approval Workflow
- This included Web UI Development, Workflow Design, Schema Design and development, and wrote several components in Java, Groovy, Python

#### **DEVELOPMENT TEAM MANAGER, EQUITY RISK MANAGEMENT IT | 04/2011 to 10/2012**

##### **ING - West Chester, PA**

- Responsible for research, development and design of financial software written in C++ used to hedge large block of variable annuities
- My responsibilities included developing new models, optimizing performance of simulations, implementing new hedging and reserve calculations required by the business, and researching new computational platforms such as GPGPU and Cloud Computing for offloading computations
- Optimized Dynamic Annuity Factor (DAF) calculations which resulted in  $O(1)$  improvement in complexity
- Implemented Hull White 1-Factor Interest Rate Model and Economic Scenario Generation using Monte Carlo
- Designed & implemented Nested Tree Map Data Structure in order to report aggregated policyholder cash flows based on selected keys
- This allows business users to analyze future cash flows for large number of Policy Holders based on requested criteria
- GPGPU Project: Led the migration of C++ Variable Annuity hedging software to run on NVIDIA hardware using CUDA

- Efforts consist of redesigning C++ data structures and rewriting existing code base to be device friendly and computationally efficient
- Also designed and implemented threading model in order to saturate device by mapping host threads to device streams and running concurrent CUDA kernels
- GPGPU Project: Redesigned correlated random deviate generation by rewriting Mersenne Twister 19937 in CUDA and implementing polar rejection on the device
- GPGPU Project: Developed core libraries to be used in CUDA
- This includes writing CUDA Launch framework in order to provide clean separation of host & device code, implementing a simple::vector (std::vector without iterators) for device, and simple::multi\_array (slimmed down version of boost::multi\_array supporting extents & strides)
- Worked with Market Data from Bloomberg and SuperDerivatives
- Developed customizations for Asset Control product in order to generate volatility surface, interest rate curve, and correlations from daily market feeds

**VICE PRESIDENT, ENGINEERING LEAD. DEVELOPMENT ENGINEERING** | 06/2005 to 04/2011  
**Citigroup - New York, NY**

- Engineering Lead for application team that was responsible for implementing enterprise level technology solutions on Linux / Solaris / AIX platforms globally
- Responsibilities included solution architecture & design, schema development, designing and architecting automation tools, and defining global standards, and documentation of best practices
- Developed Java applications to convert configuration files into XML using JAXB
- The tools were used to parse and reconstitute files configuration files to and from XML and migrating configuration into different environments using XML representation
- Designed a strategic large-scale automated deployment solution for TIBCO applications handling 1000 deployments a month, resulting in significant cost savings
- Designed and developed Java tools to communicate to WebDAV repository, and convert the XML artifacts into a CSV file for loading into Oracle Tables
- Implemented environment build automation tools using Java / XML / Perl scripts
- Wrote UNIX shell scripts that ran on Solaris / AIX / Linux to generate cloud environments. Scripts parsed configuration and generated running environment.

**DEVELOPMENT TEAM LEAD, IDENTITY AND ACCESS MANAGEMENT** | 06/2003 to 06/2005  
**KPMG - Montvale, NJ**

- Managed all development team activities for Identity Management Services team
- Worked extensively with authentication and identity protocols
- Worked with LDAP, SecurID
- Implemented in C and C++ Windows and Linux

**INDEPENDENT CONSULTANT** | 01/2003 to 05/2003  
**Thirst Inc / Pfizer - New York, NY**

- Designed and implemented a desktop sales tool to be used by the Pfizer Sales Force
- Class relationships modeled using UML in Visio
- Front-end tool implemented as JavaScript Application in Internet Explorer
- Business logic was implemented VC++ and interfaced via COM
- ADO was used to access Sybase SQL Anywhere Database

**SENIOR PROFESSIONAL SERVICES CONSULTANT** | 10/2001 to 01/2003  
**Netegrity - Rochelle Park, NJ**

- Managed and implemented Identity Management Java Solutions for clients including Housing and Urban Development, Ernst & Young, Wagner College, KPMG
- Developed custom business rules using Java and Log4J

**CONSULTING ENGINEER, PROFESSIONAL SERVICES** | 12/2000 to 10/2001  
**nCipher - Boston, MA**

- Developed Custom Security Software for clients leveraging nCipher Hardware Security Modules (HSM)

- Wrote C Drivers to interface HSM on the Tru64 platform
- Customized OpenSSL to load/store Trusted Certificates in HSM NVRAM
- Wrote Netscape Directory Encryption Plugins leveraging Symmetric keys in HSM

#### **SENIOR DEVELOPER | 07/1998 to 12/2000**

##### **Lockstar (Startup) - Clifton, NJ**

- Developed Security Software using C++ and Python for IBM OS390, Linux, and Solaris
- Invented Secure Protocol for HTTP Transactions, US Patent App 20020023207 (Secure data transfer between a client and a back-end resource via an intermediary)
- Coordinated the security review for the security product with Counterpane Labs
- The security review consisted of protocol reviews and source code changes
- Lead team in Mainframe Integration
- This included the design and development of the RACF Authentication Library, where I managed the work of two mainframe consultants, and developed C++ library for integrating with RACF

#### **SENIOR DEVELOPER | 01/1998 to 06/1998**

##### **Syndata Technologies (Startup) - Clifton, NJ**

- Added key management interface to manage public keys on a LAN
- Added code to communicate with an LDAP server to import Public Key Certificates

#### **CHEMICAL ENGINEER, APPLIED THERMODYNAMICS GROUP | 06/1994 to 01/1998**

##### **Exxon Research & Engineering - Florham Park, NJ**

- Developed Thermodynamic models used in-flow simulations, including model development and fitting parameters using statistical regression methods against experimental data
- Researched and developed new fluid mixing model (for both viscosity and thermal conductivity); implemented as part of the reactor simulator tool with that was the new Exxon standard
- Model predicted viscosity and thermal conductivity better than any known methods
- Converted Equations of State Models from FORTRAN to C, interfaced models to Excel

## EDUCATION

##### **Rutgers, The State University of NJ - New Brunswick, NJ**

###### **M.S.**

Mathematics with Option in Mathematical Finance, **05/2011**

Master's These: Pricing Convertible Bonds with Stochastic Interest Rates using Finite Differences. Priced convertible bonds using an ADI scheme which solved a multidimensional partial differential equation containing interest rate, credit, and equity market risk. Implemented in C++ and called from Excel to price callable options on convertible bonds.

##### **New York University (Courant) - New York, NY**

###### **M.S.**

Mathematics, **05/1999**

##### **Rutgers University - New Brunswick, NJ**

###### **B.S.**

Chemical Engineering, **05/1994**

##### **Rutgers University - New Brunswick, NJ**

###### **B.S.**

Mathematics & Computer Science, **05/1994**

## LANGUAGES

##### **Polish:**



Native/ Bilingual